

5CS037- Concepts and Technologies of AI

Coursework 2: Report Writing

Ethics of AI

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Submission	: 21/01/2024

Abstract

This report provides insight into the ethical components involved in AI emphasizing the societal aspects and morality of AI. The development and deployment of Artificial Intelligence (AI) have an impact on individuals, communities, and societies which makes it even more crucial for AI to have certain regulations and rules such that it would become inclusive as well as ethical. This report provides the ethical aspects involved in AI, the dilemmas and questions related to moral AI and its social good, the initiatives taken in the field of Ethical AI, and the framework and principal required in the development of ethical AI and its uses in an ethical way.

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1. Introduction

The field that deals with the things that are bad or good morally is ethics (Singer, 2023). Thus the term ethics of AI judges the moral principality of good or bad in terms of AI. People have been interacting with AI most of the time because our cell phones and computers have been loaded with AI systems be it in maps, search engines, video recommendations, or in social media. Most people and in most fields people are merely unaware that they are being influenced by AI in their day-to-day lives. AI suggests the options in these fields that it finds fitting for an individual based on their previous choices and information which means there are many other options that people are simply unaware of. Thus, making people pigeonholed in thinking and deciding what sort of a person they are (Canca, 2021).

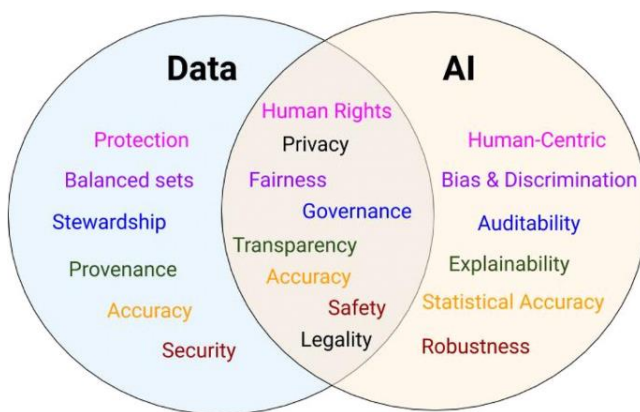


Figure 1 Factors in AI Ethics (Kazim & Koshiyama, 2021)

In general, ethical AI comprises factors like bias avoidance, protecting user privacy and data privacy, and decreasing risks to the environment. The responsible bodies for ethical AI include government, academics, non-profit organizations, private companies, and intergovernmental entities (Coursera, 2024). Ethics in AI is necessary to develop healthy AI-driven techs, for user privacy, fewer risks to the environment, and less discrimination in users (Wizata, 2022).

2. Ethical Dilemmas and Moral Questions / AI and Social Good



Figure 2 AI for Social Good (AI World School, 2024)

2.1 Environmental Impact of Training Large AI Models

Predicting natural disasters even before they occur, tracking the diseases during their spread time to eliminate them, enhancing people's lives who are especially able, etc. are some of the questions to consider when associating with AI and Social Good (Google AI, 2024). Largely training AI models that are mainly deep learning models require extremely large data processing powers thus making an impact on energy consumption. Training these models require huge electricity consumption which in turn emits large amounts of carbon which may come from non-renewable resources. Likewise, the components for hardware parts also require the mining of materials which can contribute to ecological wastes and pollution if not handled properly. Similarly, discarding unused and damaged hardware components can cause hazards too.

2.2 Ethical Considerations and Sustainability with International Cooperation and Governance Frameworks

All the wastes produced need to be recycled and the development in efficient algorithms which preserve energy should be done. Likewise, using renewable energy sources for data centers should be considered. Optimizing the hardware components can also be another way to sustainability. Being transparent in the field of data utilization and the impacts caused by AI projects in the environment can be ethical. It is also necessary to establish regulatory bodies to guide deployment of AI models which ensures sustainability and is also ethical.

Global standards for AI development are necessary to establish to address ethical and environmental concerns. Regulations that can harmonize AI community should be

considered. Other factors include information-sharing platform across countries, policy coordination among countries and governments and multi-stakeholder engagement.

2.3 AI and Social Inequalities:

If the models are trained on biased data then the output may lead to a biased result. Thus, addressing biases in datasets and algorithms can aid in creating a fair AI model which in turn mitigates social inequality caused by them. Equal access to AI technologies can also be a factor that bridges the gap between social groups. The developers and organizations must be held accountable for any discrimination shown by the model by frequent audits, addressing unintended consequences, etc. Development teams and stakeholders too can come from diverse communities which can also be a contributing factor to address the issue.

3. Initiatives in the Field of Ethical AI

EU AI Act: This act can also be considered as the first rule that is imposed on AI. EU has proposed the AI Act, a complete framework that is used to regulate applications related to AI. Many factors such as human oversight, accountability, and transparency are considered in this act. Strict penalties for not following the act are also issued thus tackling any harms and concerns caused by misuse of AI (News European Parliament, 2023).

Canadian AI Strategy: AI strategy based on responsible AI development has been pushed forward in Canada through the Pan-Canadian AI Strategy which is to address excellence in research, implications to society as well as development of the workforce. This initiative ensures that AI techs are following ethical principles (Digital Trade & Data Governance Hub, 2024).

India's National AI Strategy: This strategy of India ensures the ethical components of AI as well as covers the legal framework required to supervise AI. This is responsible for both the growth of AI and research on AI meanwhile addressing both societal needs and ethics (Ministry of Electronics & Information Technology, Government of India, 2024).

Partnership on AI (PAI), OECD AI Principles, UNESCO's Recommendation on AI, and the G7 nations have taken international initiatives for risk management of AI systems and their transparency too through data protection, less bias, and cyber security management (Cohen, 2021).

4. Generic Framework/Principle for the development and use of AI

Vivid frameworks and principles through national and international initiatives suggest different key factors for the use as well as the development of AI technologies that minimize the potential harms and hazards caused by AI.

Transparency and Explain Ability: This framework includes actions of the AI system that should be made clear to the user in real-time as a part of (SA) situation awareness. The logical process, reasoning, and factors through which the recommendation system is based and through which the AI system makes decisions should be provided in a backward-looking manner (National Academies of Sciences, Engineering, and Medicine, 2022).

Non-discrimination and Fairness: Ethical AI models and systems and frameworks should be fair and non-discriminative and other processes that involve using personal data to discriminate between people can lead to a violation of the fairness principle. Unbiasedness in algorithms and datasets is a must for this framework (ico., 2024).

Accountability and Human Oversight: Accountability ensures that the organization and developers are held responsible for the impacts of their AI systems and establishing human oversight in the framework requires human oversight in the decision-making processes of AI (European Commission, 2021).

Inclusivity and Accessibility: Ethical AI frameworks stand for the development of such AI technologies that are inclusive and can be accessed by all, regardless of social, economic, cultural, and geographical factors which lead to diverse perspectives. Even the obstacles with disabilities can be mitigated (AIContentfy, 2023).

Other frameworks and factors to be considered can be international collaboration and standards, societal impact assessment, data privacy and security, continuous education, and research.

5. Discussion and Conclusion

There must be balance between AI innovation and responsibilities that AI should handle. Building ethical AI require different approaches which not only should consider technical aspects in AI development but also should be for the society thus considering ethical and societal impact and aspects. The developers and organizations building ethical AI should be conscious about the impacts their models could create on the

society and community and thus should follow certain guidelines from international and national policies and standards. The AI should be fair, transparent, collaborative and inclusive thus making it as an ethical AI and it should not be considered as a one-time effort but continuous improvement in the journey to adapt the technologies to human world following societal aims, aspirations and values.

References

AI World School, 2024. *artificial-intelligence-for-social-good/*. [Online]

Available at: <https://aiworldschool.com/research/artificial-intelligence-for-social-good/>
[Accessed 21 01 2024].

AIContentfy, 2023. *impact-of-ai-on-content-accessibility-and-*

inclusivity#:~:text=In%20recent%20years%2C%20Artificial%20Intelligence,information%20and%20ideas%20more%20easily. [Online]

Available at: <https://aicontentfy.com/en/impact-of-ai-on-content-accessibility-and-inclusivity#:~:text=In%20recent%20years%2C%20Artificial%20Intelligence,information%20and%20ideas%20more%20easily>.
[Accessed 21 01 2024].

Canca, C., 2021. *What is an Ethical Artificial Intelligence? Mozilla Explains*, s.l.: Mozilla.

Cohen, P., 2021.

#:~:text=The%20G7%20nations%20encourage%20organizations,security%20standards%2C%20and%20bias%20minimization.. [Online]

Available at: <https://www.pearlcohen.com/global-initiatives-in-the-regulation-of-ai/#:~:text=The%20G7%20nations%20encourage%20organizations,security%20standards%2C%20and%20bias%20minimization>.
[Accessed 21 01 2024].

Coursera, 2024. *ai-ethics*. [Online]

Available at: <https://www.coursera.org/articles/ai-ethics>
[Accessed 21 01 2024].

Digital Trade & Data Governance Hub, 2024. *canada-ai-strategy/*. [Online]

Available at: <https://datagovhub.elliott.gwu.edu/canada-ai-strategy/>
[Accessed 21 01 2024].

European Commission, 2021. *1.html#:~:text=Human%20oversight%20helps%20ensuring%20that,-command%20(HIC)%20approach*. [Online]

Available at: [https://ec.europa.eu/futurium/en/ai-alliance-consultation/guidelines/1.html#:~:text=Human%20oversight%20helps%20ensuring%20that,%20command%20\(HIC\)%20approach](https://ec.europa.eu/futurium/en/ai-alliance-consultation/guidelines/1.html#:~:text=Human%20oversight%20helps%20ensuring%20that,%20command%20(HIC)%20approach).
[Accessed 21 01 2024].

Google AI, 2024. *social-good/*. [Online]

Available at: <https://ai.google/responsibility/social-good/>
[Accessed 21 01 2024].

ico., 2024.

#:~:text=Fairness%20means%20you%20should%20handle,will%20violate%20the%20fairness%20principle.. [Online]

Available at: <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/artificial-intelligence/guidance-on-ai-and-data-protection/how-do-we-ensure-fairness-in-ai/what-about-fairness-bias-and-discrimination/#:~:text=Fairness%20means%20you%20should%20handle,wi>
[Accessed 21 01 2024].

Kazim, E. & Koshiyama, A., 2021. *The interrelation between data and AI ethics in the context of impact assessments*, s.l.: AI Ethics.

Ministry of Electronics & Information Technology, Government of India, 2024. *meity.gov.in/*. [Online]
Available at: <https://www.meity.gov.in/>
[Accessed 21 01 2024].

National Academies of Sciences, Engineering, and Medicine, 2022. *Human-AI Teaming: State-of-the-Art and Research Needs..* Washington, DC: The National Academies Press.

News European Parliament, 2023. *eu-ai-act-first-regulation-on-artificial-intelligence*. [Online]
Available at: <https://www.europarl.europa.eu/news/en/headlines/society/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence>
[Accessed 21 01 2024].

Singer, P., 2023. *ethics-philosophy*. [Online]
Available at: <https://www.britannica.com/topic/ethics-philosophy>
[Accessed 21 01 2024].

Wizata, 2022. *the-importance-of-ethics-in-ai*. [Online]
Available at: <https://www.wizata.com/knowledge-base/the-importance-of-ethics-in-ai>
[Accessed 21 01 2024].